

SURPLUS-TO-SHELTER | UNIFIED SYSTEM ARCHITECTURE

Real-Time Food Rescue Routing & Redistribution Engine * Full-Stack Production Blueprint

1. PRESENTATION LAYER (React 18 SPA + Vite + Tailwind CSS + Leaflet Engine)

Donor Portal

- Post Surplus Food (Qty, Expiry, Veg)
- GPS Coordinates (Auto-Detect)
- Full Post CRUD (Edit / Cancel Post)
- Active Delivery Real-Time Tracking

NGO / Shelter

- Match Radar Feed (Candidate ranking)
- Capacity Slider (Live Intake limits)
- 1-Click Accept / Justified Reject
- Compatibility Score Breakdown Modal

Driver Hub

- Nearest Radar (Haversine distance km)
- Turn-by-Turn FSM Delivery Tracking
- Web Audio Dispatch Alert Chime
- Proximity Countdown & Arrival Detector

Admin Command

- Leaflet Real-Time Rescue Mesh Map
- Carbon & Meal ESG Automated Audit
- Role Access & RBAC Permissions
- System Dispatch Queue Observability

2. API GATEWAY & TRANSPORT (Express 4)

- **Unified Serving:** Port 5000 + SPA Wildcard Fallback app.get("**")
- **Security Guard:** JWT 'protect' Middleware + RBAC Role Authorization
- **RESTful Endpoints:** Full CRUD across /api/donations, /api/ngos, /api/drivers
- **Socket.IO Mesh:** Isolated rooms (user_{id}, feed), live event push
- **Dual Architecture:** Persistent server.js (Render) & serverless api/index.js (Vercel)
- **Public Impact API:** Unauthenticated live metrics (/api/dashboard/public-stats)
- **Connection Pooling:** SSL/TLS encrypted Mongoose pool with serverless caching

3. DOMAIN LOGIC: MATCHING ALGORITHM & DELIVERY FSM

Multi-Factor Matching Scoring Formula:

$$\text{Score} = (0.35 * S_{\text{dist}}) + (0.30 * S_{\text{cap}}) + (0.20 * S_{\text{urg}}) + (0.15 * S_{\text{exp}})$$

Distance <= 15km Haversine | Intake >= Meals | Safe Transit < Usable Expiry Buffer

Delivery State Machine (FSM):

[POSTED] → [MATCHED] → [ASSIGNED]

→ [PICKUP_STARTED] → [PICKED_UP] → [DELIVERED]

Turnkey state advancement with proximity arrival detection and shelter capacity updates

ESG Carbon Model: 1 Meal = 0.42 kg Food * 1 kg Food = 2.5 kg CO2e Avoided

Automated ESG social calculation on verified delivery (eliminates landfill methane decomposition)

4. DATA PERSISTENCE & MULTI-TARGET DEPLOYMENT INFRASTRUCTURE

MongoDB Atlas Replica Set

- Cluster: cluster0.i2iv3xy.mongodb.net
- Database: surplus_to_shelter (3-node geo-replica)
- Serverless conn caching: mongoose.readyState >= 1
- Strict Zero Dummy Data policy (Manual seedData.js)
- 8-second cloud handshake timeout with safe fallback
- SSL/TLS encrypted connection pool (SRV protocol)

Mongoose Schemas & Indexes

- Users & Roles: DONOR, NGO, DRIVER, ADMIN
- Profiles: DonorProfile, NGOProfile, DriverProfile
- Donation: Category, quantity, expiryTime, safety, match
- Delivery: Driver, NGO, audit timestamps, status
- 2dsphere Geospatial Index on [longitude, latitude]
- Compound query indexes on: { status: 1, createdAt: -1 }

Multi-Target Production Deployment

- VERCEL DEPLOYMENT (vercel.json):
 - Global Edge CDN frontend from client/dist
 - Node.js Serverless Function entry at api/index.js
- RENDER DEPLOYMENT (render.yaml):
 - Persistent Web Service (Node.js + WebSockets)
- ALL-IN-ONE BUILD:
 - npm run build (compiles client) && npm start (port 5000)